Advanced Financial Management - Case

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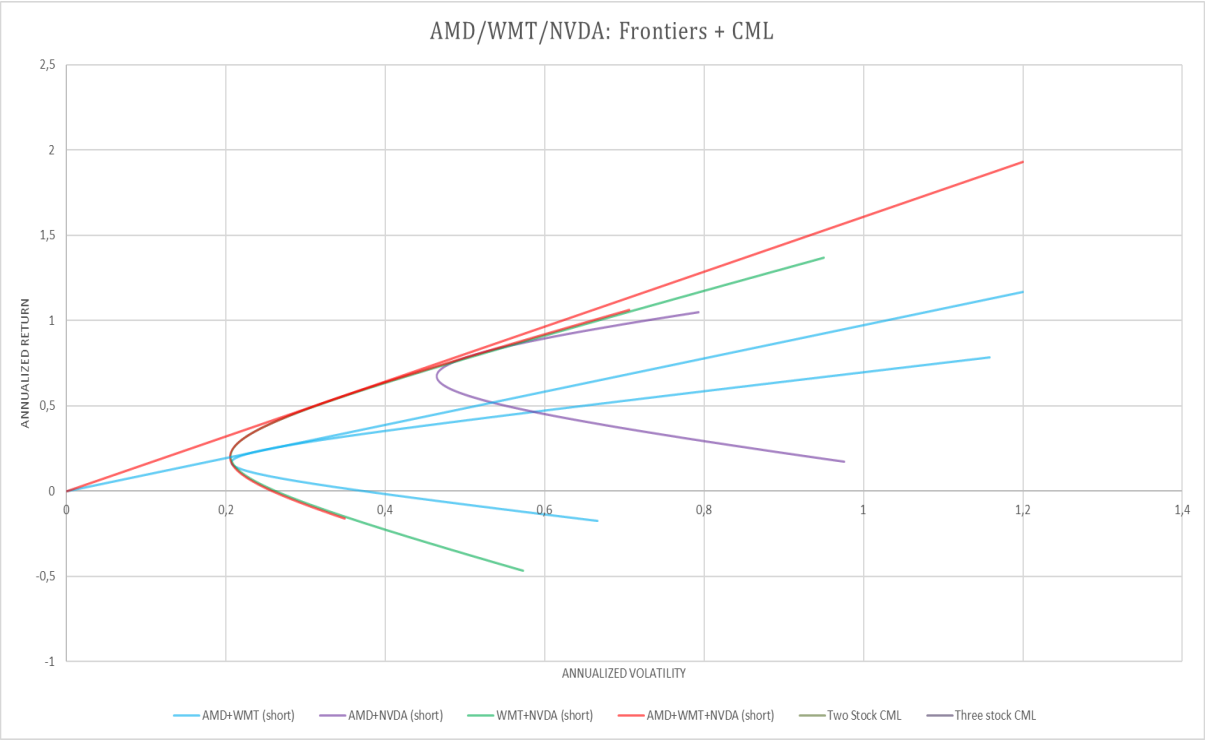
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Link to Google Docs:

https://docs.google.com/document/d/1rmWiwCoKJk2Ba0knzE8c1tKzCbEoCfVnmHFNt8PLBSo/edit?tab=t.0

Page 1: Efficient frontier

The graph below illustrates the efficient frontier and capital allocation line for a portfolio consisting of three stocks: AMD, WMT, and NVDA. The horizontal axis represents risk, measured by annualized standard deviation, while the vertical axis represents annualized returns. The efficient frontiers of the two-stock combinations (AMD-WMT, NVDA-AMD, and NVDA-WMT) are shown by the light blue, purple and green curves, respectively. Each curve demonstrates the trade-off between risk and return, which can be adjusted by varying the portfolio weights of the two assets. The three-stock efficient frontier is represented by the red curve. The three-efficient frontier is very close to the WMT-NVDA frontier, which means that adding AMD does not offer major diversification benefits. A possible explanation for this could be the high correlation between AMD and NVDA of 0.76 (Macroaxis, 2025). This may be due to their shared position as major tech semiconductor firms. WMT and NVDA on the other hand only have a correlation of 0.48 (Macroaxis LLC, 2025). The capital market lines (CML) are depicted as two straight lines. The blue line corresponds to a two-stock portfolio of WMT and NVDA, while the red line corresponds to the three-stock frontier. Both lines start from the risk-free rate and extend upward as risk increases. The slope of each CML reflects the Sharpe ratio of the optimal portfolio: the steeper the slope, the greater the return per unit of risk. This indicates that the three-stock portfolio offers a more efficient mix of risk and return.



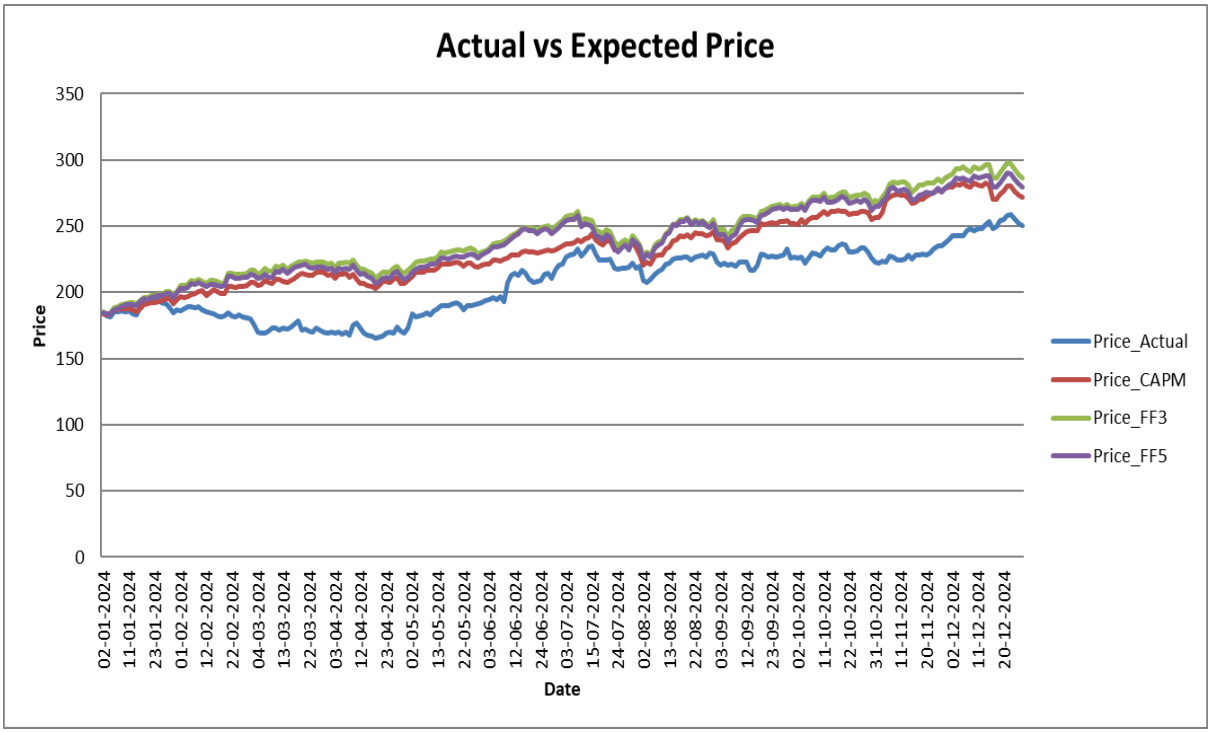
Page 2: Beta estimation

The graph below shows how Apple's (AAPL) beta fluctuated over these eight years. In 2018, the U.S. economy experienced strong growth, while tensions from the U.S.-China trade war significantly increased AAPL's volatility. Because Apple relies heavily on production in China, tariff concerns increased in COGS and reduced profit, amplifying its impact from the systemic risk, leading to a rise in beta (Gurman, 2018). This rise is highlighted in the graph below by the first red line. During the COVID-19 pandemic, APPL's beta declined heavily, highlighted by the second red line. As people spent more time at home, APPL's service revenue increased significantly, accounting for approximately 20% of the company's total revenue in 2020 (Downey, 2021). Investors perceived AAPL as a low-risk company due to its substantial cash reserves and its reduced vulnerability to pandemic-related disruptions, resulting in a lower beta during this period. In 2022, APPL's beta rose again (see third red line) amid an economic downturn and high inflation in the U.S., which forced the Federal Reserve to raise interest rates (Budget, 2023). These rate hikes further discounted AAPL's future cash flows, reducing its NPV and increasing its systemic risk sensitivity. By the end of 2024, market volatility had declined, and the government began to reduce the interest rates (*Federal Reserve Issues FOMC Statement*, n.d.). Consequently, systemic risk decreased, and AAPL's returns became less sensitive towards market fluctuations, leading to a lower beta. Additionally, the company's stable earnings and strong cash reserves contributed to reduced sensitivity during this period.



Page 3: Factor models

The graph below compares Apple's (AAPL) actual stock price evolution in 2024 with estimated expected prices using three different models. CAPM, shown by the red line, is based solely on the market factor. The Fama-French 3-Factor Model (FF3), represented by the green line, incorporates size, market, and value factors. The Fama-French 5-Factor Model (FF5), shown by the purple line, adds profitability and investment factors to the three-factor model. All expected price paths were computed using historical data. For our graph, AAPL was selected as it produced interesting results. All three models overestimated the stocks's return; however, CAPM provided the closest estimate among the three. This outcome can be attributed to the relatively high market beta across all three models (ranging from 1.16 to 1.23), as shown in the factor loading table, included in appendix 4. This indicates that APPL's performance is primarily driven by overall market movements. The remaining factors exhibit small to negative coefficients, suggesting that they introduced additional noise rather than adding explanatory power. Consequently, the market beta alone outperformed the additional factors - SMB, HML, CMA, and RMW - because these factors were neither strong nor stable enough during the evaluation window.



Page 4: Project Finance

The financial projection table below outlines the following: revenue growth (10% in the first two years and 6% thereafter); cost of goods sold (COGS) (at 60% of revenue); and linear depreciation ($1.7 bn over 10 years). Net working capital (NWC) is projected at 5% of revenue. Free cash flow (FCF), which reflects the cash available to both equity and debt holders, is calculated by adding back depreciation as a non-cash expense. The cost of equity for Apple (AAPL) is estimated using the CAPM model, with the risk-free rate obtained from FRED database and the market risk premium (MRP) from Damodaran's 2024 implied equity risk premium. The beta is derived from median beta calculation for AAPL on page 2. For the discount rate calculation, the WACC method has been applied. The pre-tax cost of debt is based on AAPL's bond yield reported by FINRA, while debt and equity figures are market values. Case 1 represents AAPL's current debt-equity ratio for which the NPV is $107.02 million. Case 2 assumes equity-only financing, which increases the discount rate and lowers the NPV to $91.62 million. The discount rate is higher as the risk is borne by equity shareholders demanding a higher return and as there is no debt, there are no tax shields that lower the firm's WACC. Case 3 illustrates a scenario where the level of debt is doubled (higher leverage), but the WACC (discount rate) is lower as there is a tax shield, so the NPV increases to $121.69 million. These cases are projected in the left-bottom table.

AAPL Financial Projection Table ($ millions)

Item / Year	0	1	2	3	4	5	6	7	8	9	10
Income Statement											
Sales (Revenues)	0	900	990.00	1,089.00	1,154.34	1,223.60	1,297.02	1,374.84	1,457.33	1,544.77	1,637.45
Less:											
COGS (60%)		540	594	653.4	692.60	734.16	778.21	824.90	874.40	926.86	982.47
Gross Profit(In Millions)		**360**	**396**	**435.6**	**461.74**	**489.44**	**518.81**	**549.93**	**582.93**	**617.91**	**654.98**
Less:											
SGA Expenses(constant)		150.00	150.00	150.00	150.00	150.00	150.00	150.00	150.00	150.00	150.00
Depreciation(in Millions)		170.00	170.00	170.00	170.00	170.00	170.00	170.00	170.00	170.00	170.00
EBIT		**40.00**	**76.00**	**115.60**	**141.74**	**169.44**	**198.81**	**229.93**	**262.93**	**297.91**	**334.98**
Taxes (25%)		25.00%	25.00%	25.00%	25.00%	25.00%	25.00%	25.00%	25.00%	25.00%	25.00%
Net Income		**30.00**	**57.00**	**86.70**	**106.30**	**127.08**	**149.10**	**172.45**	**197.20**	**223.43**	**251.24**

Cash Flow From Assets	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
Add Back: Depreciation		170.00	170.00	170.00	170.00	170.00	170.00	170.00	170.00	170.00	170.00
Subtract: CapEx	(1,700)										
NWC (5% of sales)	0	45	50	54	58	61	65	69	73	77	82
Change in NWC		45	5	5	3	3	4	4	4	4	5
Project Free Cash Flow (FCF) (in Mil)	(1,700)	**155.00**	**222.50**	**251.75**	**273.04**	**293.62**	**315.43**	**338.56**	**363.07**	**389.06**	**416.60**

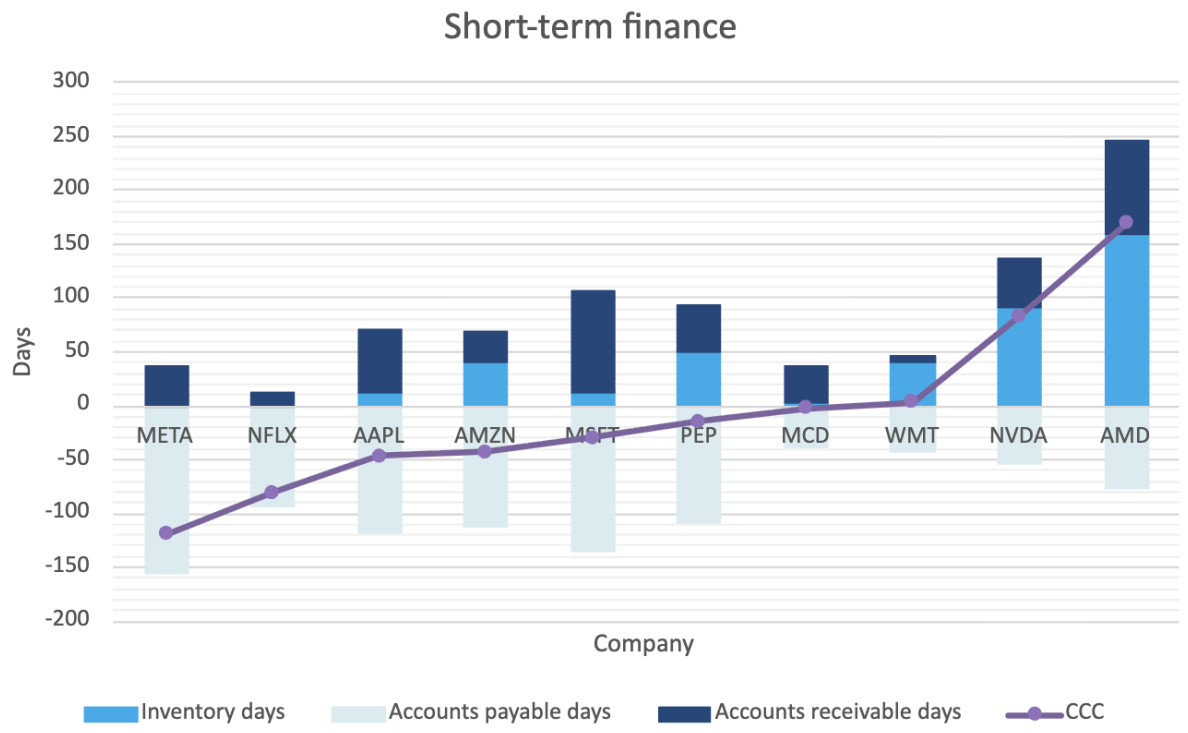
Year	FCF	Discount Factor	PV (Rwacc= 9.13%)	Discount Factor	PV (Rwacc=Re= 9.30%)	Discount Factor	PV (Rwacc= 8.97%)
0	(1,700)	1	(1,700)	1	(1,700)	1	(1,700)
1	155	0.916	142.036	0.915	141.816	0.918	142.243
2	222.5	0.839717769	186.837	0.837	186.259	0.842	187.383
3	251.75	0.769484746	193.718	0.766	192.820	0.773	194.568
4	273.035	0.705125932	192.524	0.701	191.335	0.709	193.651
5	293.6171	0.646150014	189.721	0.641	188.257	0.651	191.110
6	315.4341	0.592106774	186.771	0.587	185.043	0.597	188.413
7	338.5602	0.542583648	183.697	0.537	181.717	0.548	185.583
8	363.0738	0.497202579	180.521	0.491	178.299	0.503	182.641
9	389.0582	0.455617131	177.262	0.449	174.808	0.462	179.605
10	416.6017	0.417509841	173.935	0.411	171.263	0.424	176.492
		NPV (in mil)	**USD 107.02**	**NPV (in mil)**	**USD 91.62**	**NPV (in mil)**	**USD 121.69**

AAPL CAPM Calculation	
Risk free Rate (Rf) (FRED Database)	4.08%
Market risk Premium (Damordaran Implied ERP 2024) (Rm-Rf)	4.14%
Beta	1.26
Cost of equity	**9.30%**

WACC Calculation				
Debt	1,19,059.00		Cost of debt (pre tax) (YTM FINRA)	5%
Equity	37,85,545.55		Cost of debt (after tax)	3.75%
Debt	0.03		Cost of equity	**9.30%**
Equity	0.97			
Tax Rate	25.00%		**Wacc**	**9.13%**

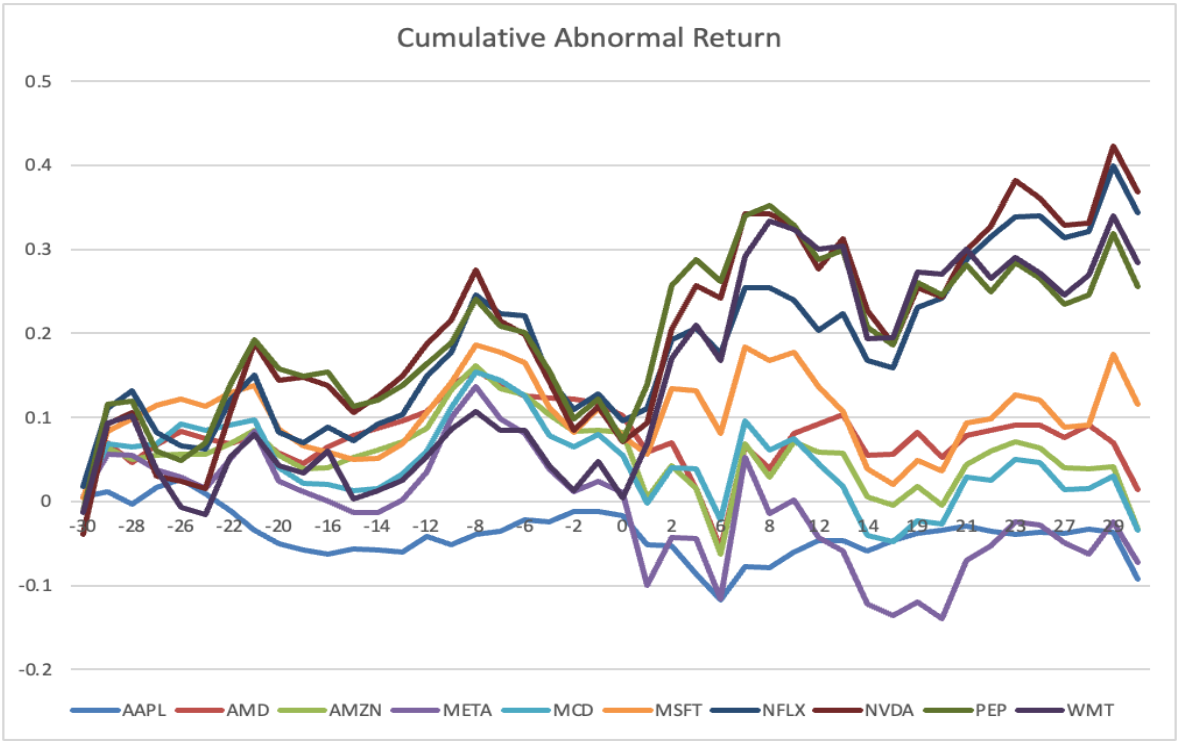
Page 5: Short-term finance

This section presents a graph illustrating short-term finance indicators for the year 2024. The graph displays *inventory days*, *accounts payable days* and *accounts receivable days* stacked in a bar for each of the ten companies. The horizontal axis represents the companies, sorted from lowest to highest CCC. The CCC is calculated by adding the inventory days and accounts receivable days, and then subtracting accounts payable days. For detailed calculations and underlying data, refer to Appendix 5. It is immediately apparent that most companies exhibit a negative CCC, which is desirable for most companies. A negative CCC indicates financial efficiency, suggesting the firm finances its business operations with consumer payments (Triple Whale, 2024). META and Netflix stand out with particularly low (i.e., negative) CCCs, primarily due to not having any inventory. Additionally, their accounts receivable days are lower relative to their accounts payable days, indicating META and Netflix collect consumer payments faster than paying their suppliers. In contrast, Nvidia has a high CCC, largely driven by high inventory days. Nvidia should aim to accelerate its inventory turnover to shorten its CCC, improving their operational efficiency by better predicting future sales in the semiconductor sector. AMD has the highest CCC at 170 days. To reduce its CCC, AMD should slow down its production and lower its inventory levels. This would reduce inventory days, and in turn, lower the CCC. A lower CCC indicates faster conversion of inventory into profit, which is financially desirable and attractive to investors, as it is an indicator of financial health (Turkingston, 2025).



Page 6: Event study Liberation Day

The graph below illustrates the cumulative abnormal returns (CAR) for ten stocks surrounding Liberation Day, April 2nd 2025 - the date on which President Donald Trump announced new tariffs. Prior to the event (day 0, the 'shock'), several market movements reflected anticipation to the event, like China's retaliatory tariffs against U.S. products on day -22 (PBS, 2025). Among the observed stocks, Apple (AAPL) was hit the hardest. Given Apple's reliance on manufacturing in China and subsequent imports to the U.S., the tariffs were expected to raise costs of goods sold (COGS) and compress profit margins (Leswing, 2025). As a result, investors expected a negative outcome from the event, and AAPL's CAR went negative both before and after the announcement. In contrast, NVIDIA (NVDA), a U.S.-based leading AI and semiconductor company, benefited from the tariff protection. Its CAR rose as investors anticipated stronger relative performance. Most stocks exhibited similar trends before and after the announcement, with the notable exception of META. Initially, the market anticipated a positive outcome for META, driven by expectations of higher social media engagement and tech optimism in general. However, following the shock, investors reassessed META's valuation. Trump's tariffs introduced broader economic uncertainty, likely to suppress advertising expenditures, which is META's main source of revenue (Poletti, 2025). All three stocks discussed are in the tech sector; however, the impact of the event differed between these stocks. To better measure this, variables like imported COGS and sales origin could explain which companies were hit hard.



Appendix 1: References

A timeline of Trump's tariff actions so far. (2025, April 3). PBS News. https://www.pbs.org/newshour/economy/a-timeline-of-trumps-tariff-actions-so-far

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Poletti, T. (2025, April 22). *Meta may have a $7 billion problem in China*. MarketWatch. https://www.marketwatch.com/story/meta-may-have-a-7-billion-problem-in-china-14e51886

Turkington, L. (2025, March 28). *Negative Cash Conversion Cycle: Definition, benefits and key Strategies*. B2BE. https://www.b2be.com/blog/negative-cash-conversion-cycle/

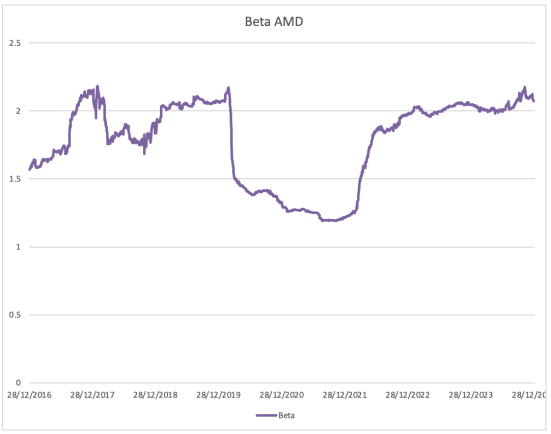
Appendix 2: Page 1

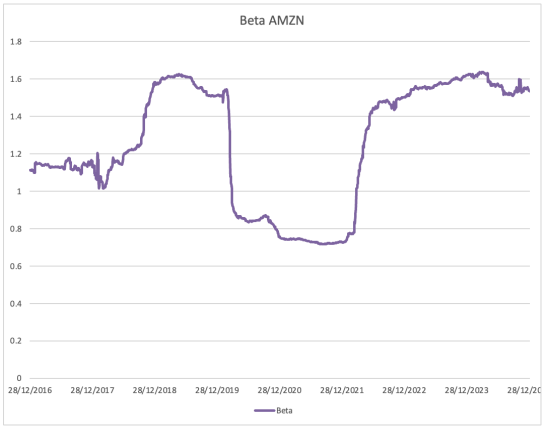
In order to estimate a 2 stock efficient frontier we first calculated the portfolio return using the formula $E[R_p] = w_A * E[R_A] + W_B * E[R_B] + W_C * E[R_C]$ were w_A, W_B and W_C are the weights of stock A, B or C and $E[R_A]$, $E[R_B]$ are the expected annualized returns of stock A, B or C. Then we calculated the portfolio volatility and used the formula $\sigma_p = SQRT\ (W_A^2\sigma_A^2 + W_B^2\sigma_B^2 + 2 * w_A * W_B * COV$, were σ_A^2 is the annualized volatility to stock A, and COV is the covariance. In order to create the three stock efficient frontier we also used the return of the three stocks portfolio and created a variance-covariance matrix. Afterwards we used an excel function that randomizes a series of numbers in order to randomize weights for the portfolio. Then, we used 'what if' analysis in Excel, and we randomized weights for the portfolio 2000 times, plotting this gave the curved red line.

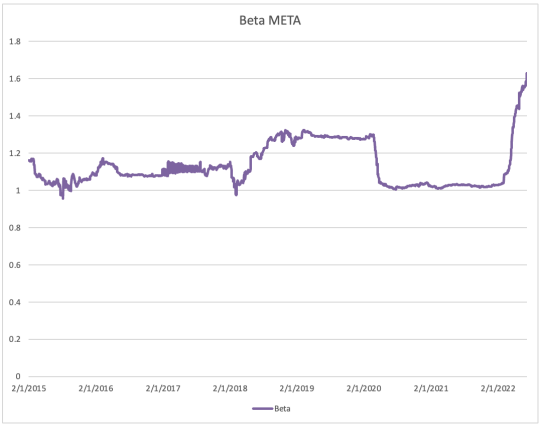
Appendix 3: Page 2

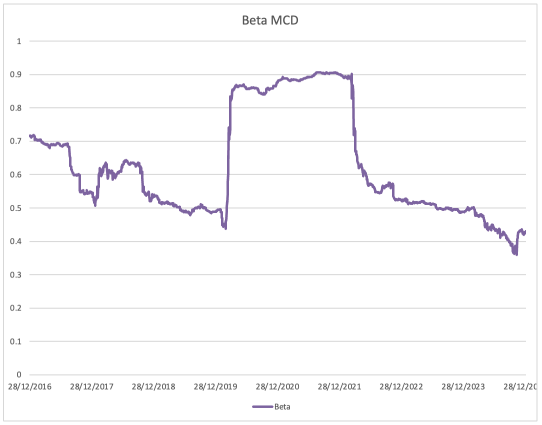
Beta is estimated using formula *cov(stock return, market return)/var(market return)*, rolling window 2 years on total 10 years data as requested. Both stock return and market (S&P500) return data was gathered from CRSP. Below are the graphs of all 10 stocks we chose: AAPL, AMD, AMZN, META, MCD, MSFT, NFLX, NVDA, PEP, and WMT.

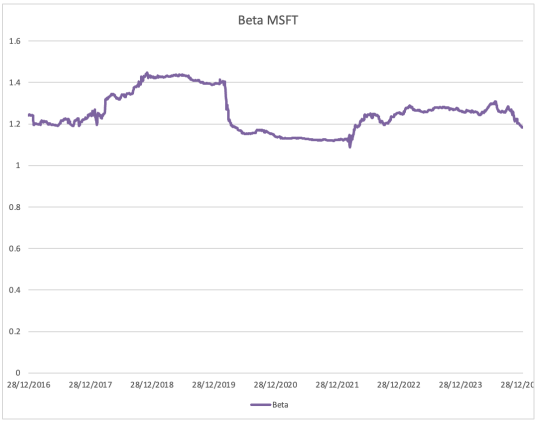


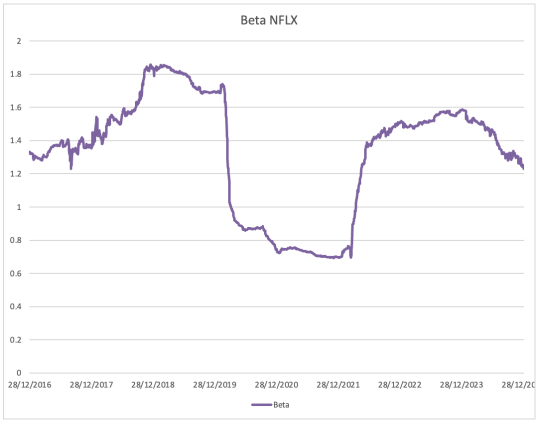


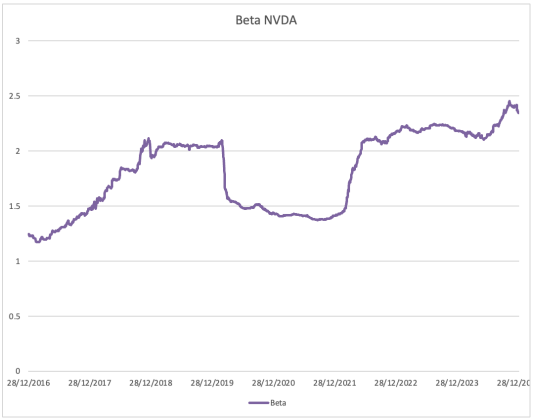


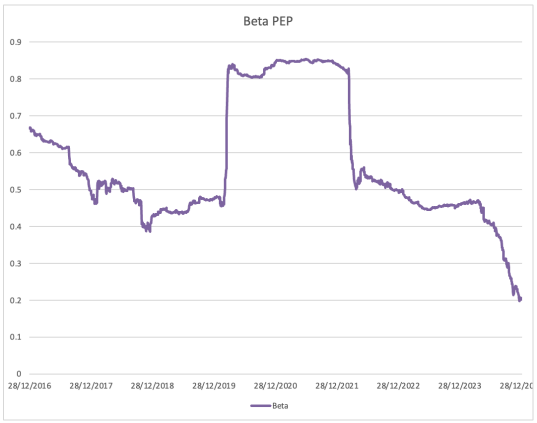







Beta NFLX
2
1.8
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
0
28/12/2016
28/12/2017
28/12/2018
28/12/2019
28/12/2020
28/12/2021
28/12/2022
28/12/2023
Beta


Beta NVDA
3
2.5
2
1.5
1
0.5
0
28/12/2016
28/12/2017
28/12/2018
28/12/2019
28/12/2020
28/12/2021
28/12/2022
28/12/2023
Beta


Beta PEP
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
28/12/2016
28/12/2017
28/12/2018
28/12/2019
28/12/2020
28/12/2021
28/12/2022
28/12/2023
Beta


Beta WMT
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
28/12/2016
28/12/2017
28/12/2018
28/12/2019
28/12/2020
28/12/2021
28/12/2022
28/12/2023
Beta

Appendix 4: Page 3

Factor loading table used in computing the graph on page 3:

Model	Factor	Loading (beta)
CAPM	MKT	1.168793238
FF3	MKT	1.193779866
FF3	SMB	-0.358974399
FF3	HML	-0.33577998
FF5	MKT	1.238952163
FF5	SMB	-0.21677451
FF5	HML	-0.536444245
FF5	RMW	0.451886133
FF5	CMA	0.26947021

Appendix 5: Page 4

We have chosen AAPL to study the effects of different combinations of debt and equity financing. The financial projection is calculated using the information given in the problem statement where the project outflow/investment for year 0 is $1.7billion. The revenue is $900 million for the first year and increases for 10% for 2 years and 6% after that for 7 years. SGA expenses remain constant at $150 million and COGS are 60% of revenue. Depreciation is $1.7billion depreciated linearly over 10 years ($170 million). Tax rate is 25%.

Projection table calculation-

Gross profit = Revenue - COGS

EBIT = Gross profit - SGA expenses - Depreciation

Net income = EBIT*(1-tax rate)

FCF = Net income + Depreciation - Capex - Change in NWC

CAPM calculation -

$Re = Rf + \beta*(Rm-Rf)$

Risk free rate (*Rf*) source : https://fred.stlouisfed.org/series/DGS10

Beta source : Page 2 beta estimation calculation of AAPL

Market risk premium source :
https://pages.stern.nyu.edu/~adamodar/New_Home_Page/datafile/histimpl.html

WACC calculation -

$Rwacc = E/D+E * (Re) + D/D+E * (Rd)$

Equity and Debt figures source(Apple website) :
https://investor.apple.com/sec-filings/sec-filings-details/default.aspx?FilingId=17933082

Cost of debt (Rd) for AAPL source :
https://www.finra.org/finra-data/fixed-income/bond?cusip=037833AL4&bondType=CA

NPV calculation-

$NPV = -outflow + \sum FV/(1+Rwacc)^n$

Case 3: WACC calculation when debt to equity ratio is doubled:

WACC Calculation			
Tax rate	25.00%	Rd (pre tax) (FINRA)	5%
Debt	238118.00	Rd (After tax)	3.75%
Equity	3785545.55	Cost of equity	**9.30%**
Debt	0.06		
Equity	0.94	**WACC**	**8.97%**

Appendix 6: Page 5

The table below shows the underlying values for the calculation of the CCC for each of the ten companies.

Company	Inventory days	Accounts payable days	Accounts receivable days	CCC
META	0	156,3572991	36,79155446	-119,566
NFLX	0	92,59562086	12,2709867	-80,3246
AAPL	12,29323727	118,7804598	59,37859782	-47,1086
AMZN	40,654525	111,2438609	27,51979359	-43,0695
MSFT	12,63342203	135,2072939	93,29819437	-29,2757
PEP	50,54664723	107,5161391	42,01787619	-14,9516
MCD	1,954715495	38,30200335	34,29619985	-2,05109
WMT	40,73374534	42,25256626	5,030518293	3,511697
NVDA	91,09878148	53,42461413	46,23999019	83,91416
AMD	158,7469812	76,87855787	87,18382781	169,0523

The inventory days is calculated by: inventory days = (average inventory/GOGS)*365

The accounts payable days is calculated by: accounts payable days = (average accounts payable/COGS)*365

The accounts receivable days is calculated by: accounts receivable days = (average accounts receivable/sales)*365

The cash conversion cycle (CCC) is computed by: CCC = inventory days + accounts receivable days - accounts payable days

Appendix 7: Page 6

For this event study, first we estimate the return using the CAPM model.

$Ra = Rf + [\beta a * (Rm - Rf)]$

Return on market is calculated from the S&P500 price, source: https://fred.stlouisfed.org/series/SP500#. The risk free rate is from the US treasury bill 1 month from FRED, translated to daily, since the rate provided is annual. Source: https://fred.stlouisfed.org/graph/?id=DTB3. We subtract Rm and Rf to get MRP daily.

Beta is the beta calculated using all data from question number 2.

By knowing Rf, beta, and Rm, we can estimate the expected return on stocks daily.

To get the abnormal return, we subtract actual return on the expected return calculated using CAPM previously. Then we can get the daily abnormal return.

Return on stocks is calculated from the stock's price. Since CRSP does not provide data for 2025, we download it from NASDAQ https://www.nasdaq.com/market-activity/stocks/wmt/historical?page=1&rows_per_page=10&timeline=y1.

Adding all the daily abnormal return from day -30 to day 30, we get the cumulative abnormal return. Applying this method to all stocks, we can plot the graph and do comparison.